



✦ JOB READY PROGRAM

DevOps & Multi Cloud + GenAI Program

A comprehensive, industry-aligned training program covering DevOps fundamentals, Azure, AWS, GCP cloud architecture, and cutting-edge GenAI integration — from beginner to expert.

150+
HOURS

4
TRACKS

35+
MODULES

10+
PROJECTS

BEGINNER TO EXPERT · JOB READY

Placement-Focused, Not Just Certification-Focused — Build a real portfolio of enterprise-grade projects that gets you hired.



Microsoft
Partner

PROGRAM OVERVIEW



COURSE NAME

**DevOps with MultiCloud
& GenAI**



LEVEL

Beginner to Expert



DURATION

150+ Hours

COURSE CURRICULUM

TRACK 01

Generic DevOps

11 Modules

MODULE 01

DevOps Overview

Waterfall → Agile → DevOps evolution · CI/CD · Lifecycle stages · Automation

MODULE 02

Linux OS Fundamentals

Architecture · Distributions · File permissions · SSH · Shell scripting

MODULE 03

Python Scripting

OOPs · Loops · File ops · Functions · Exception handling

MODULE 04

Git & GitHub

VCS · Branching · Merging · Rebase · Remote repositories · VS Code

MODULE 05

Maven Build Tool

Architecture · Plugins · Build web, standalone & enterprise apps

MODULE 06

Jenkins CI/CD

Pipelines · Master-Slave · Tomcat · Build triggers · Email notifications

MODULE 07

Docker & Docker Hub

Images · Volumes · Networking · Compose · Swarm · Jenkins integration

MODULE 08

Kubernetes Orchestration

Pods · Deployments · Services · Volumes · AKS/EKS/GKE · Scaling

MODULE 09

Terraform (IaC)

HCL · AWS resources · Modules · Remote state · Provisioners

MODULE 10

Ansible Automation

Architecture · Playbooks · Variables · Handlers · Galaxy roles

MODULE 11

Prometheus & Grafana

Continuous monitoring · Dashboard visualization · Alerting

TRACK 02

Microsoft Azure — Associate

10 Modules · Microsoft Certified

MODULE 01

Getting Started with Azure

Cloud overview · IaaS/PaaS · Azure CLI · PowerShell

MODULE 02

Microsoft Entra ID

Azure AD · MFA · SSPR · RBAC · Conditional policies

MODULE 03

Compute Services

VMs · Scale sets · Availability zones · Custom images

MODULE 04

Azure Networking

VNet · VPN · Load balancers · Firewall · Front Door

MODULE 05

Azure Storage

Blob · Table · Queue · File · CDN · ARM Templates

MODULE 06

PaaS & Container Services

ACI · ACR · AKS · Function App · Logic App

MODULE 07

Azure Data Platform

Azure SQL · Cosmos DB · Elastic pools · Triggers

MODULE 08

App Security

Key Vault · Managed identities · AD app management

MODULE 09

Backup & Monitoring

Recovery vault · Azure Monitor · Log Analytics · Alerts

MODULE 10

Azure DevOps

Boards · Repos · Pipelines · Web app deployments

TRACK 03

AWS Solution Architect — Associate

8 Modules

MODULE 01

Cloud Architecting & EC2

Well Architected Framework · EC2 · EBS · AMI · CloudWatch

MODULE 02

Storage Services

S3 · Versioning · Lifecycle · EFS · FSx · Cross-region replication

MODULE 03

Security & Databases

IAM · Directory Service · RDS · Multi-AZ · SSO

MODULE 04

Networking

VPC · ELB · Auto Scaling · CloudFront · Route 53

MODULE 05

Decoupled Architecture

SNS · Lambda · Event-driven design

MODULE 06

Disaster Recovery

DR strategies · Storage Gateway · Hybrid migration

MODULE 07

Additional AWS Services

CodePipeline · ECR/ECS · Systems Manager · Security Hub

MODULE 08

Capstone Project

High availability architecture · Dynamic website on RDS

TRACK 04

GCP Architect — Self Learning

15 Modules

MODULES 01–03

Cloud Foundations & IAM

GCP overview · Regions · Console · IAM roles · Service accounts

MODULES 04–06

Compute, Storage & LB

Compute Engine · GCS · Load balancing · Autoscaling

MODULES 07–09

Networking & Databases

VPC · Hybrid network · Cloud SQL · High availability

MODULES 10–12

Advanced Databases

Cloud Spanner · Firestore · Bigtable · Replication

MODULES 13–14

Serverless & Observability

App Engine · Cloud Functions · Cloud Run · Monitoring

MODULE 15

GKE & Kubernetes

Microservices · Docker · K8s clusters · Pods · YAML

TRACK 05

GenAI Integration with DevOps

6 Modules

MODULE 01

Introduction to AI/ML

Supervised · Unsupervised · Reinforcement · Deep learning · Neural networks

MODULE 02

Generative AI Fundamentals

LLMs · GPT · Claude · Gemini · RAG · Prompt engineering

MODULE 03

Why GenAI + DevOps?

Self-healing systems · AI-assisted debugging · Intelligent CI/CD

MODULE 04

GenAI Tools for DevOps

GitHub Copilot · CodeWhisperer · LLM-based IaC · Log anomaly detection

MODULE 05

MLOps with GenAI

MLOps lifecycle · AWS/Azure/GCP ML · Model monitoring · Drift

MODULE 06

AI Assistants for CloudOps

ChatOps bots · LLM + Jenkins/GitLab CI · Prompt-driven agents

HANDS-ON PROJECTS

Project 01

Dynamic Blog Site on AWS

Secure, highly available website using AWS Storage and Internet Gateways.

Project 02

Deploy Web Apps on Azure

Containerized app deployed via Azure Container Registry to Azure App Service.

Project 03

E-Commerce App on AKS

Scalable microservices e-commerce platform on Azure Kubernetes Service.

Project 04

Dynamic Infrastructure with Terraform & Ansible

Highly reliable dynamic infrastructure management on Azure.

Project 05

Dedicated VPC on Multi-Cloud

Web app deployment in dedicated VPCs on AWS/Azure/GCP using Terraform.

Project 06

Azure Serverless Architecture

Serverless web application using Azure Functions.

Project 07

Automated Monitoring & Alerts

Azure Monitor with proactive alert management and automated webhooks.

Project 08-10

Advanced Projects

Auto-scaling environments · Advanced networking · Cosmos DB NoSQL apps.

REAL-WORLD CASE STUDIES



CI/CD Pipeline for Web App

Jenkins Docker
Kubernetes Terraform



On-Premise to Azure Migration

Azure Terraform Ansible



IaC Implementation

Terraform Ansible Azure



Microservices Monitoring

Prometheus Grafana
Helm



Dockerize & Orchestrate with K8s

Docker Kubernetes Helm



DevSecOps Pipeline

Snyk SonarQube Jenkins



High Availability & DR

Azure LB VM Scale Set



Blue-Green Deployment

Azure DevOps App Services



Paperlive Learning

Empowering careers through industry-aligned technology education.

www.paperlive.in